To:	MTC
From:	WSP
Cc:	
Ref:	
Date:	2026-06-08
Subject:	Truck Time-of-Day Distribution Calibration

Purpose

This memo provides documentation of the newly estimated truck time-of-day distributions developed for TM1.7. A comparison with the current TM1.6 distributions is included to summarize how the updated estimates differ from the existing model assumptions.

Data Sources

Observed data are derived from a combination of Caltrans 2018 and BATA 2023 datasets. Caltrans data provide detailed vehicle classification, while BATA contributes bridge-level counts.

Data Processing and Assumptions

Exclusion of BATA Small Truck Observations

Small truck observations from the BATA 2023 dataset are excluded from the analysis. These observations are based solely on the 2-axle classification, which combines both very small (2-axle, 4-tire) and small (2-axle, 6-tire) trucks. Because the BATA data do not distinguish between tire configurations, it is not possible to separate these two vehicle types.

To avoid introducing bias into the estimation of time-of-day distributions, all small truck observations from the BATA 2023 source are removed. This exclusion has a limited impact on the overall dataset, as BATA 2023 provides counts for only seven bridges. The majority of observations are sourced from Caltrans data, where very small and small trucks can be identified and estimated separately.

Results

TM1.6 Distributions

Truck Type	EA	AM	MD	PM	EV
Very Small	2.35%	0.00%	60.80%	19.80%	17.05%
Small	7.65%	24.40%	37.10%	21.80%	9.05%
Medium	6.65%	29.30%	39.35%	17.30%	7.40%
Large	14.30%	23.20%	33.15%	17.50%	11.85%

Observed Distributions

Truck Type	EA	AM	MD	PM	EV
Very Small	9.16%	23.79%	30.23%	23.89%	12.93%
Small	9.67%	27.13%	34.42%	19.32%	9.46%
Medium	9.53%	32.52%	34.59%	15.00%	8.36%
Large	13.40%	25.20%	33.41%	11.63%	16.36%

Difference (Observed - TM1.6)

Truck Type	EA	AM	MD	PM	EV
Very Small	6.81%	23.79%	-30.57%	4.09%	-4.12%
Small	2.02%	2.73%	-2.68%	-2.48%	0.41%
Medium	2.88%	3.22%	-4.76%	-2.30%	0.96%
Large	-0.90%	2.00%	0.26%	-5.87%	4.51%